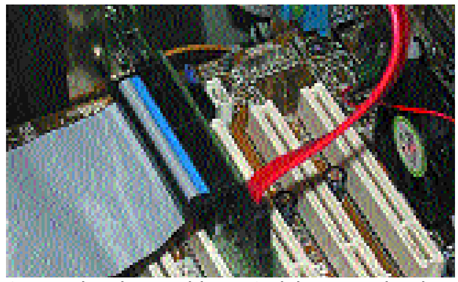
Serial ATA: Hit or Miss?

With the upcoming ATA/133 standard pushing the limits of the maximum transfer rates supported by the parallel ATA interface, the next step to revolutionise hard disk interfaces will be Serial ATA. It offers greater speed, easier fabrication and application.

The Serial ATA interface is to ATA what FireWire is to SCSI—the entire approach is to convert the parallel standard that is used by the existing ATA interfaces to a serial format (where one bit is transferred at a time) so that higher data transfer frequencies can be supported. Also, this approach greatly simplifies the fabrication of cables and connecting wires that are used by the interface since there are fewer signals to deal with, facilitating a less cluttered cabinet. This translates into easier airflow within the system and subsequently, more efficient cooling. In fact, Serial ATA uses just four signal pins (there are a total of seven pins; the others are used for transmitting overhead commands) that actually carry the data and include the power and ground pins—significantly lesser than the 80 lines that are used in today's ATA/100 and ATA/133 cables.



Conventional IDE cables vs Serial ATA: notice the difference (flat vs rounded)

This standard will support transfer rates of up to 1.5 Gigabits per second, which translates into about 150 MBps. Also, since this is a point-to-point protocol, there will be no place for master and slave devices on a single cable and therefore, bandwidth will not need to be shared between devices. Another very big advantage with this standard is that it is backward compatible with the existing parallel ATA standard at the driver level. Therefore, all existing applications in which the parallel ATA standard is used can be fully extended to support Serial ATA.

Specifications: *150 MBps maximum transfer rate (300/600 MBps envisioned for the future), hot-plugging capability, two power saving modes: partial and slumber, overlapping (commands), tagged command queuing, seven-wire data cable, connectors measure just 8 mm wide.*

excellent read burst speed of 80 MBps, which is the highest speed a drive can achieve across an ATA/100 interface. The drive that really disappointed here was the 80 GB Seagate ST380021A. It scored a very paltry 56.7 MBps in this test. This was surprising considering the fact that this drive sported a 7,200-rpm spindle speed.

In the CPU utilisation test, the drive that broke barriers was Western Digital's 200 GB monster (the WD2000), which logged a stunning 0.5 per cent of CPU utilisation. The Maxtor 80 GB drive also scored over the other drives in this category with a good CPU utilisation score of 3.7 per cent. If you use space consuming and CPU-intensive applications such as Photoshop or AutoCAD, the drives with least CPU utilisation should be your pick, especially when you are running these applications together and would be transferring data between them.

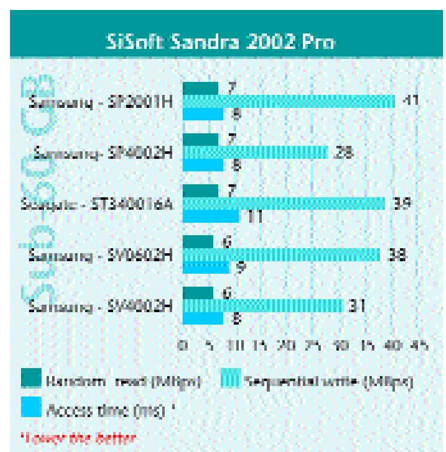
SiSoft Sandra 2002 Pro: This is a synthetic benchmark that evaluates the sequential and random read and write transfer rates and the access time of the disk.

Up to 60 GB capacity: In the SiSoft Sandra 2002 Pro sequential read speed test, the 5,400-rpm drive that took the lead was the 60 GB Samsung SV0602H. It logged a very respectable score of 38 MBps. The highest scores in this test for the 7,200-rpm drives was a tie between the two Samsung hard disks, the SP2001H and the SP4002H, both scoring 41 MBps.

In the random read tests, the performance variations were not that wide. The scores ranged between 5 MBps and 7 MBps—in practical terms, the speed advantage offered by the upper value of this variation is insignificant.

Over to the write tests. When dealing with sequential data, the drive that did well in the 5,400-rpm range was the Samsung SV0602H with a very respectable score of 38 MBps. The 20 GB Maxtor 541DX managed to rake up a score of just 12 MBps for sequential writes.

The 7,200-rpm drive that did well in this category was once again the Samsung SP2001H, with a very respectable score of 41 MBps. Surprisingly, the other 7,200-rpm entry from Samsung, the 40 GB SP4002H, trailed in this test with a score of just 28 MBps.



In the random write test, there was a reasonably large variation in score with the 40 GB Seagate ST340810A logging a very respectable 7 MBps in spite of being a 5,400-rpm drive. The drive that did badly here was again the 20 GB Maxtor 541DX with a score of just 3.1 MBps.

> **info** A typical hard disk drive today can be completely assembled from its components in about 11 seconds—the entire process is, naturally, automated

The access time determines the speed with which the drive can locate the date you need, especially if it is scattered all over the hard disk. The fastest access time was logged by four of the five Samsung drives—the SP2001H, SP4002H, SV2001H and SV4002H each logged a respectable 8 ms, while the Samsung SV0602H logged 9 ms. The drives that fared particularly poorly in this test was the 20 GB Maxtor 541DX and the Seagate ST340016A—both logged 11 ms.

Over 60 GB capacity: The fastest hard disk in the sequential read test was the Western Digital